

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**First/Second Semester of B. Tech. (CE/IT/EC/CL/ME/EE) Theory Examination****April-May 2018****ME141/ME101.01/ME101 Engineering Graphics****Date: 08/05/2018, Tuesday****Time: 01:30 p.m. to 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and Section II must be attempted in separate drawing sheets.
3. Make suitable assumptions and draw neat figures wherever is required.
4. Use of scientific calculator is allowed.
5. Figures drawn in the question paper are not to the scale.
6. Dimensions shown in the figure are in mm.
7. Show all the necessary dimensions in the solution.
8. Figures to right indicate the full marks.

SECTION – I**Q - 1 Do as directed.**

1. _____ are used to fix the drawing paper on the drawing board. **[01]**
a) Drawing pins b) Drawing pens c) Drawing pencils d) None of these
2. 1 foot = _____ inches. **[01]**
a) 1 b) 6 c) 12 d) 24
3. In first-angle projection method, the left-hand side view is drawn at the right side of the front view. True or False (T/F)? **[01]**
4. What is the full form of BIS? **[01]**
a) Bureau of Indian Standards b) Bureau of International Standards c) Basic of Indian Standards d) Basic of International Standards
5. Isometric drawing represents a 3D object of 2D views. True or False (T/F)? **[01]**
6. The isometric drawing of a circle is an ellipse. True or False (T/F)? **[01]**
7. Enlist the applications of CAD Modeling Software. **[01]**

Q – 2.a Explain first angle projection system and third angle projection system with necessary sketches and discuss the difference between first angle projection system and third angle projection system. **[08]**

Q – 2.b A cone has a diameter of base 40 mm and height of axis 70 mm long, is resting on its base on HP. Draw the isometric drawing of the cone. **[08]**

OR

Q – 2.a **Figure 1** shows a pictorial view of an object. Draw the following views: (1) Front view, and (2) Top view. Dimensions shown in the figure are in mm. (**Figure 1** is on Page 3) **[08]**

Q – 2.b **Figure 2** shows the orthographic projections of an object in first angle projection system. Draw its isometric view. Dimensions shown in the figure are in mm. (**Figure 2** is on Page 3) **[08]**

Q – 3.a Distinguish between engineering drawing and engineering graphics. **[02]**

Q – 3.b Draw an ellipse by using any method. Consider major axis of an ellipse as 100 mm and minor axis of an ellipse as 80 mm. Draw normal and tangent at any point on the curve. **[06]**

OR

Q – 3.b Draw and explain how to bisect a line and how to bisect an angle. **[06]**

Q – 3.c Give the overview of AutoCAD. **[04]**

SECTION – II

Q - 4 Do as directed.

1. What do you mean by projection of points? **[01]**
2. The intersection of HP and VP is an XY line. True or False (T/F)? **[01]**
3. When a line is parallel to HP and perpendicular to VP, the true length of line is visible in top view. True or False (T/F)? **[01]**
4. A plane is perpendicular to HP and VP then it becomes parallel to PP. True or False (T/F)? **[01]**
5. When a point is above HP, in which of the following quadrant it lies? **[01]**
 a) 2nd quadrant b) 3rd quadrant c) 4th quadrant d) None of these
6. A straight line is a shortest distance between two points. True or False (T/F)? **[01]**
7. Define development of solid. **[01]**

Q – 5.a Classify the solids with necessary sketches. **[09]**

OR

Q – 5.a A cylinder of 50 mm diameter of base and 80 mm height of axis is resting on the H.P on its base. It is cut by an A.I.P. in such a way that the true shape of the section is an ellipse of the largest possible major axis. Draw the sectional plan and true shape. Find the inclination of the sectional plane. **[09]**

Q – 5.b A cube of 40 mm side is resting on HP on its base. Draw the development of cube. **[06]**

Q – 6.a A point P is 50 mm from both the reference planes. Draw its projection in all possible positions. **[05]**

Q – 6.b A regular hexagonal plate of 30 mm side is resting on one of its corners in VP. The diagonal through that corner is inclined at 40° to VP. (1) The front of that diagonal is inclined to HP by 45° . (2) The diagonal is inclined to HP by 45° . Draw the projections of hexagonal plate. **[08]**

OR

Q – 6.a A 60 mm long line AB has one end A in VP and other end B is 15 mm above HP. Draw the projections of the line when the line is inclined 30° to HP and 60° to VP. **[07]**

Q – 6.b ABCD is a rhombus of diagonal AC as 60 mm and BD as 40 mm long. A corner A of a rhombus is in the HP. The plane is inclined to the HP such that the plan appears as a square. The plan of diagonal AC makes an angle of 20° to the VP. **[06]**

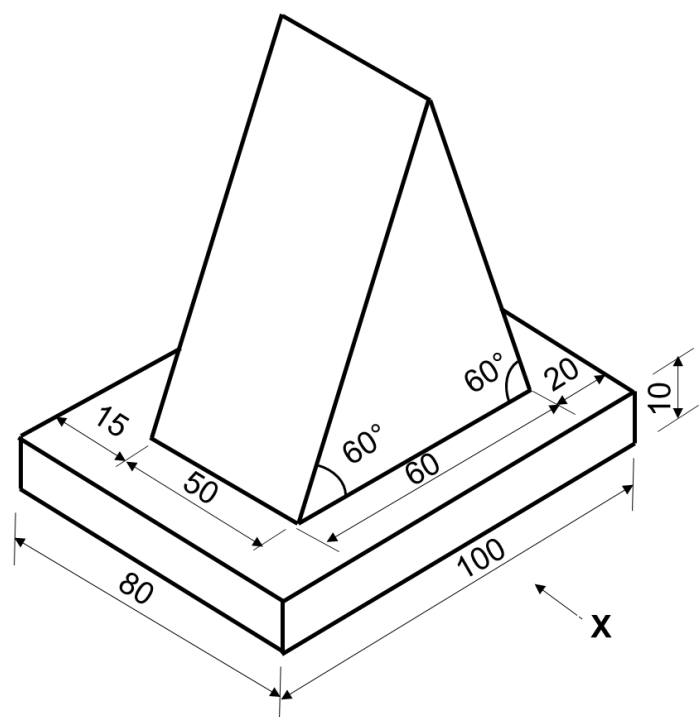


Figure 1

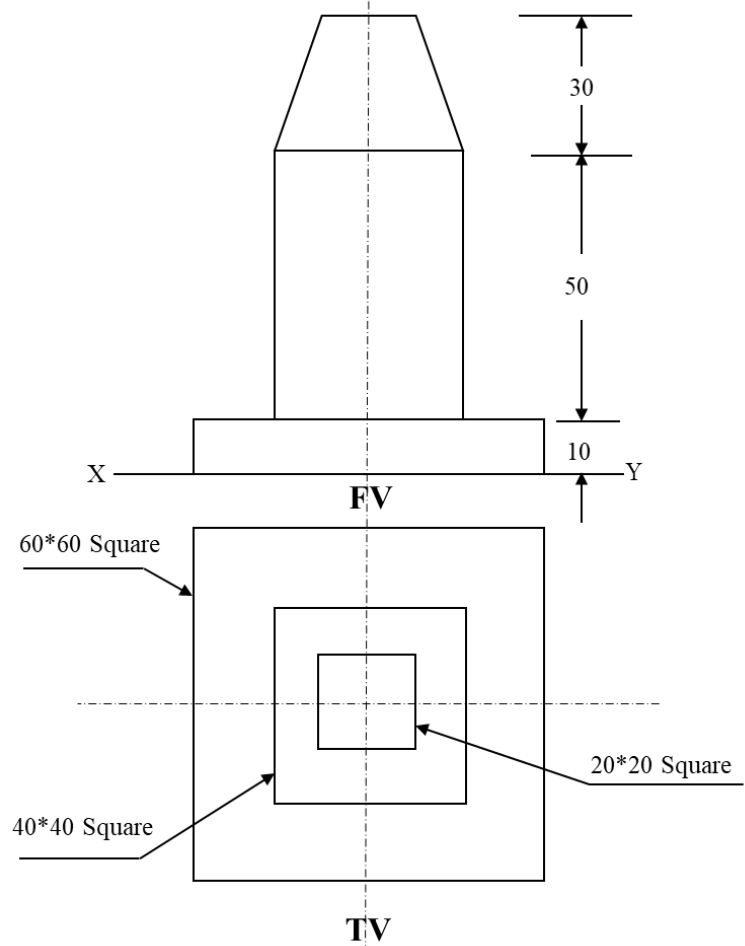


Figure 2
